

Cheat Sheet: Installation and Configuration of OpenVAS

Estimated duration: 3 minutes

Welcome! This cheat sheet is designed to help security professionals and system administrators quickly set up and operate OpenVAS (Greenbone Vulnerability Management) for effective vulnerability scanning. It outlines essential commands, configurations, and troubleshooting tips to ensure smooth scanning of your network and systems.

OpenVAS is a powerful open-source vulnerability scanner. It is part of the Greenbone Vulnerability Management (GVM) solution and provides comprehensive scanning and reporting capabilities. This cheat sheet walks you through the key steps from installation to scan report generation using both the command line and web interface.

Use this sheet to:

- Set up and validate the GVM environment
- Configure targets and scanning tasks
- Perform authenticated scans
- Access and export scan reports
- Resolve common setup issues

System setup and service management

Command	Description
sudo apt update && sudo apt install openvas -y	Installs OpenVAS on Debian-based systems
sudo gvm-setup	Initializes GVM services, generates certificates, and downloads initial feeds
sudo gvm-start	Starts OpenVAS-related daemons (gvmd, gsad, openvas-scanner)
sudo gvm-check-setup	Verifies setup status, including feed status and service health
sudo greenbone-nvt-sync	Manually synchronizes NVTs (vulnerability tests)
sudo journalctl -u gvmd	Checks logs for issues with the Greenbone Manager Daemon

Web interface access (GSA)

URL	Description
https://127.0.0.1:9392	Default web access to the Greenbone UI

Creating a target in GSA

1. Navigate to **Configuration** → **Targets**
2. Click **Create Target**
3. Fill in:
 - **Name:** For example, Internal Web Server
 - **Hosts:** For example, 192.168.56.101
 - **Alive Test:** ICMP/TCP/ARP
 - **Port List:** OpenVAS Default
4. Click **Save**

Creating a scan task

1. Go to **Scans** → **Tasks**
2. Click **Create Task**
3. Configure:
 - **Target:** Choose a previously created target
 - **Scan Config:** "Full and fast"
4. Start the task manually or schedule it

Adding credentials (for authenticated scans)

1. Navigate to **Configuration** → **Credentials**
2. Click **Create Credential**
3. Choose:
 - Type: SSH Password or SMB

Viewing and exporting reports

Location	Description
Scans → Reports	View completed scans, filter by severity
Export options	PDF, XML, HTML, TXT
Severity scale	Based on CVSS: Low (0–3.9), Medium (4–6.9), High (7–8.9), Critical (9–10)

Common errors and fixes

Error	Cause	Fix
"No NVTs found"	Feed not synced	Run greenbone-nvt-sync, restart services
"503 Service Unavailable" (GSA)	gvmd not running or DB connection failed	Restart gvmd, verify PostgreSQL
Scan stuck at 0%	Incorrect target IP or firewall blocking traffic	Verify network, update port lists

Tips for success

- Always sync feeds before scanning new targets
- Use credentialed scans for deeper vulnerability discovery
- Schedule scans during off-peak hours to avoid network slowdowns
- Use tags to group hosts by environment (for example, DEV, PRO)

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